English

Ha-VIS eCon 3000 Fast Ethernet Basic PoE / PoE+

HARTING

5 to 8 Ethernet ports RJ45, FOC, and PoE / PoE+
Unmanaged Plug & Play Ethernet switches for DIN rail assembly in control cabinets
Commercial temp.: 0°C ... +55°C / Industrial temp.: -40°C ... +70°C

Slim | +70 -40 | Energy Efficient Ethernet | PoE+

- Energy supply of up to 4 terminal devices via PoE+ (137 watts) in accordance with IEEE 802.3at
- Fast Ethernet Non-Blocking Switch architecture in accordance with IEEE 802.3
- Variants with industrial temperature range of -40°C ... +70°C
- Surge protection and reverse polarity protection
- Minimum energy consumption due to energy-efficient Ethernet
- Optimised DIN rail bracket

Target markets

Machinery & Robotics	Automation	Industrial network infrastructures
Wind Energy, Solar Energy	Transportation	Shipbuilding

General Description

The Ha-VIS eCon 3000 Fast Ethernet PoE family of unmanaged Ethernet switches is equipped with up to 8 Fast Ethernet ports and allow for cost-efficient and quick expansion and/or reconstruction of network infrastructures. The switches work as power sourcing equipment (PSE) and are capable of simultaneously providing the full PoE output of 34.2 watts on up to four ports. The slender design of the switches enables an extremely high packing density on the DIN rail. The selection includes various combinations of variants with RJ45 and fibre optic ports. Automatic detection of the transmission rate (auto-negotiation) and of the wiring of the twisted pair data cable (auto-polarity and auto-MDI(X)) allow for simple plug & play. All variants are available with the temperature ranges "Industrial" and "Commercial".

Specification

Switch Features

Housing width	**25 mm**
Number of ports	5, 6, 7, 8
Switching technology	Store and Forward
Supported standards	IEEE 802.3
Frame size	1522 bytes
MAC table size	1k entries
Packet buffer size	448 kbit
Non-blocking	Yes
Quality of Service	Yes
Energy-efficient Ethernet	Yes
PROFINET-compatible	Yes
EthernetIP-compatible	Yes

Power supply

Nominal voltage	24 VDC ⎓	48 VDC ⎓	54 VDC ⎓
Permissible voltage range	9 VDC ... 60 VDC ⎓		
Surge protection	Yes		
Reverse polarity proof	Yes		
Starting current	3.20 A	6.40 A	7.20 A
Overcurrent protection at input	Yes (12 A)		
Max. power consumption @ 24 VDC	1.92 W ... 3.12 W		
Conductor cross-section	0.08 mm^2 ... 2.5 mm^2 (28 AWG ... 12 AWG)		
Type of connection	3-pole, pluggable screwed contact		
Pinout	+ / - / ⏚		
Supply circuit (according to 60950)	SELV (circuit breaker 10 A)		

Ethernet ports 10BASE-Te / 100BASE-TX

Type of connection	RJ45
Auto-negotiation	Yes
Auto-polarity	Yes
Auto-MDI(X)	Yes
Transfer conditions	Twisted pair
Transfer speed	10 / 100 Mbit/s
Transfer length	100 m (Twisted Pair, Cat 5)

Ethernet ports 100BASE-FX

Type of fibre	**Multi-mode (MM)**	**Single-mode (SM)**
Type of connection	SC Duplex	
Transfer conditions	FOC	
Wavelength	1310 nm	
Transfer speed	100 Mbit/s	
Transfer length	2 km	15 km
Output power	-20 dBm ... -14 dBm	-15 dBm ... -8 dBm
Input sensitivity	≤ -30 dBm	≤ -32 dBm

Ambient conditions

Commercial temperature range	0°C ... +55°C
Industrial temperature range	-40°C ... +70°C
Storage temperature range	-40°C ... +85°C
Relative humidity (operation)	0% ... 95% (non-condensing)
Relative humidity (storage and transport)	0% ... 95% (non-condensing)
Air pressure	2000 m (795 hPa)

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PoE

Number of PoE ports	4	
Standard	IEEE 802.3af / IEEE 802.3at	
PoE type	PSE	
Supported mode	Alternative A	
Power supply PSE	48 VDC ⎓	54 VDC ⎓
Maximum power consumption PSE	1380 mA	2380 mA
Maximum current (PoE / PoE+)	375 mA	638 mA
Maximum output power PSE	15.4 W per port 61.6 W total	34.2 W per port 136.8 W total
Supported cabling	See 802.3at, section 33.1.4	
PoE pinout	Alternative A, MDI-X (1/2 = V-, 3/6 = V+)	

Status and diagnostic displays

Power ("Pwr") ⏻ illuminated green	Supply voltage is applied
Link/Activity ("L/A") off	No link
Link/Activity ("L/A") illuminated green	Link is active
Link/Activity ("L/A") flashes green	Link is active and data is transferred
Link speed ("Spd") off	10 Mbit/s
Link speed ("Spd") illuminated yellow	100 Mbit/s
PoE status off	PoE is inactive / low voltage
PoE status illuminated green	Voltage in PoE range
PoE status illuminated blue	Voltage in PoE+ range
PoE status illuminated red	Fault

Housing

Housing width	**25 mm**
Dimensions H x W x D (without pluggable screwed contact and holding bracket)	142 mm x 25 mm x 107.5 mm
Weight	480 g ... 508 g
Type of installation	35 mm DIN rail acc. to EN 60 715
Material hoods/housings	Anodised aluminium / powder-coated steel sheet
Protection class (with plugged screwed contact)	IP30
Protection class	III

Approvals (in preparation)
CE (FCC CFR 47 Part 15, cUL US 508 listed, DNV, GL, ABS, NK)

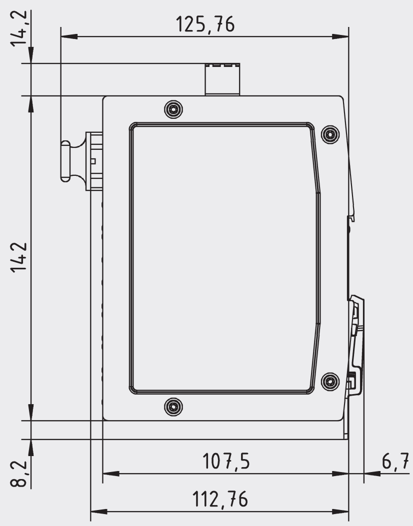
EMC and environmental conditions
EMC interference immunity (EN 61000-6-1, 61 000-6-2 55024)
Electrostatic discharge (ESD) EN 61 000-4-2
Electromagnetic field EN 61 000-4-3
Rapid transients (burst) EN 61 000-4-4
Surge voltages EN 61 000-4-5
Conducted interference voltages EN 61 000-4-6

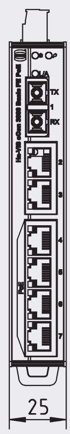
EMC interference emission (EN 61000-6-4, EN 55 022, FCC CFR 47 Part 15)

Mechanical stability (EN 60721-3)
IEC 60068-2-6 Vibration
IEC 60068-2-6 Resonance search
IEC 60068-2-27 Shock test

Scope of delivery
- Pluggable screwed contact for power supply
- Installation instructions

Drawings





Specifications / order information

Ports / order information

RJ45	SFP	SC	Housing width	Power consumption @ 24 VDC without PoE	MTBF in million h	Commercial temp.: 0°C ... +55°C Switch	Commercial temp.: 0°C ... +55°C Order no.	Industrial temp.: -40°C ... +70°C Switch	Industrial temp.: -40°C ... +70°C Order no.
4	-	1x MM (2 km)	25 mm	110 mA	1.19	Ha-VIS eCon 3041B-AD-P	24030041130	Ha-VIS eCon 3041BT-AD-P	24030041120
4	-	1x SM (15 km)	25 mm	110 mA	1.19	Ha-VIS eCon 3041B-AF-P	24030041230	Ha-VIS eCon 3041BT-AF-P	24030041220
4	-	2x MM (2 km)	25 mm	130 mA	1.17	Ha-VIS eCon 3042B-AD-P	24030042130	Ha-VIS eCon 3042BT-AD-P	24030042120
4	-	2x SM (15 km)	25 mm	130 mA	1.17	Ha-VIS eCon 3042B-AF-P	24030042230	Ha-VIS eCon 3042BT-AF-P	24030042220
6	-	-	25 mm	80 mA	1.18	Ha-VIS eCon 3060B-A-P	24030060030	Ha-VIS eCon 3060BT-A-P	24030060020
6	-	1x MM (2 km)	25 mm	120 mA	1.15	Ha-VIS eCon 3061B-AD-P	24030061130	Ha-VIS eCon 3061BT-AD-P	24030061120
6	-	1x SM (15 km)	25 mm	120 mA	1.15	Ha-VIS eCon 3061B-AF-P	24030061230	Ha-VIS eCon 3061BT-AF-P	24030061220
8	-	-	25 mm	90 mA	1.14	Ha-VIS eCon 3080B-A-P	24030080030	Ha-VIS eCon 3080BT-A-P	24030080020

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